

Phu (Otis) Han

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EDUCATION

Seattle University

Master of Science in Computer Science, Data Science — GPA: 3.925/4.0

Seattle, WA

Sep 2025 – May 2027

University of Science - VNUHCM

Bachelor of Science in Computer Science, Major: Software Engineering — GPA: 3.5/4.0

HCMC, Vietnam

Oct 2020 – Oct 2024

WORK EXPERIENCE

Software Engineer

Nov 2023 – Jul 2025

Hasaki Technology

- Engineered a distributed **Order Management System** in **Go**, consuming **Kafka** event streams and coordinating background workers to process **200K+ weekly orders** with low-latency, reliable handling under high-concurrency workloads.
- Built real-time CDC pipelines connecting (**MySQL** → **Kafka** & **Elasticsearch**), eliminating manual Kafka publishing across **90% of data synchronization workflows**.
- Redesigned a legacy synchronous export engine into an event-driven pipeline using **Go**, **Kafka**, and **MinIO**, streaming **1M+ rows** per export via **PIT/search_after** pagination; reduced API response time to **<100ms** and eliminated HTTP timeout/OOM failures.
- Built **Go/Redis/Asynq** background workers automating post-order operations, including **Third-Party Logistics courier failover**, live address/COD sync, and return refund processing, saving **10 hours/week** of manual exception handling.
- Eliminated race conditions and duplicate transactions in multi-threaded order workflows using **Redis distributed locking** and optimistic concurrency control, removing a recurring source of production incidents.

PROJECTS

CareerCompass – Job-seeking system integrated with AI

Jan 2024 – Jul 2024

- Led** a **6-member** team to architect and ship a full-stack AI recruitment platform using **FastAPI** and **OpenAI API**, achieving a **9.8/10** technical evaluation score.
- Engineered a real-time **CDC pipeline** with **PostgreSQL**, **Debezium**, and **Kafka**, streaming events to **Qdrant**, **Weaviate**, and **Elasticsearch** for real-time vector indexing.
- Designed a **RAG-based** semantic search engine with LLM query expansion, using **Qdrant** vector search and **OpenAI embeddings** to achieve **95% retrieval precision**.
- Built** recruiter-facing matching tools using vector similarity scoring to rank applicants against job postings, reducing manual screening effort.
- Developed an **AI analytics assistant** using the **OpenAI Assistants API** with **MinIO** to auto-generate data visualizations from system metrics.

Retail Sales Forecasting

Sep 2025 – Dec 2025

- Processed the **55M+ row M5 Forecasting dataset** and built time-series models (**XGBoost**, **LightGBM**, **Random Forest**) with Optuna tuning and SHAP analysis, evaluated via RMSE.
- Performed exploratory data analysis using Pandas, NumPy, and Matplotlib to identify sales patterns and forecasting signals.

TECHNICAL SKILLS

Languages: Python, Go/Golang, C/C++, Java, JavaScript, C#, TypeScript, ReactJS

Backend & Architecture: Microservices, REST APIs, FastAPI, gRPC, Kafka, RabbitMQ, Docker, Protocol Buffers

Databases: PostgreSQL, MySQL, MongoDB, DynamoDB, Elasticsearch, Qdrant, Weaviate

Cloud & Network & Tools: AWS, Azure, S3, EC2, EKS, Lambda, RDS, Spark, MapReduce, Jenkins, Argo CD, Debezium, Git, GitHub Actions, Linux

Core CS: Data Structures & Algorithms, Object-Oriented Design, TCP/IP, OSI Models, System Design

Machine Learning: Scikit-Learn, PyTorch, Tensorflow, Optuna

ACHIEVEMENTS

Leader of Capstone Project: Ranked #1/Grade 9.8 — **Dean's Graduate Student Honor:** 2026